Docosahexaenoic acid from a cultured microalga inhibits cell growth and induces apoptosis by upregulating Bax/Bcl-2 ratio in human breast carcinoma...

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Docosahexaenoic acid (DHA) is an omega-3 fatty acid that comprises 22 carbons and 6 alternative double bonds in its hydrocarbon chain (22:6omega3). Previous studies have shown that DHA from fish oil controls the growth and development of different cancers; however, safety issues have been raised repeatedly about contamination of toxins in fish oil that makes it no longer a clean and safe source of the fatty acid. We investigated the cell growth inhibition of DHA from the cultured microalga Crypthecodinium cohnii (algal DHA [aDHA]) in human breast carcinoma MCF-7 cells. aDHA exhibited growth inhibition on breast cancer cells dose-dependently by 16.0% to 59.0% of the control level after 72-h incubations with 40 to 160 microM of the fatty acid. DNA flow cytometry shows that aDHA induced sub-G(1) cells, or apoptotic cells, by 64.4% to 171.3% of the control levels after incubations with 80 mM of the fatty acid for 24, 48, and 72 h. Western blot studies further show that aDHA did not modulate the expression of proapoptotic Bax protein but induced the downregulation of anti-apoptotic Bcl-2 expression time-dependently, causing increases of Bax/Bcl-2 ratio by 303.4% and 386.5% after 48- and 72-h incubations respectively with the fatty acid. Results from this study suggest that DHA from the cultured microalga is also effective in controlling cancer cell growth and that downregulation of antiapoptotic Bcl-2 is an important step in the induced apoptosis.